

Digital Filters

Script: Chapter 10 (FIR and IIR filter design, filter structures)

Learning goals

- Turn a verbal requirement (“extract the knock resonance”) into a quantitative filter specification and estimate the necessary order.
- Design linear-phase FIR filters with the window method and IIR filters with the bilinear transform (including pre-warping); realise IIR filters as biquad cascades.
- Judge the effects of finite word length: coefficient quantisation, scaling, overflow, saturation; Q15 arithmetic on an 8-bit controller.
- Run the knock band-pass sample by sample in real time on the ESP32 (float) and on the Arduino Uno (fixed point) and compare FIR and IIR in terms of cost and delay.

Background and conventions

The knock sensor signal of the virtual engine contains the knock resonance at $f_1 = 6.5$ kHz (plus a weaker mode at 10.5 kHz), valve impacts at 8 kHz, a low-frequency combustion vibration and broadband noise. The first block of every knock detector is a band-pass filter around f_1 . On this sheet we use

sampling rate	$f_s = 25$ kHz ($T_s = 40$ μ s)
pass band	6.0 kHz to 7.0 kHz, loss at most $A_p = 1$ dB
stop bands	$f \leq 5.0$ kHz and $f \geq 8.0$ kHz, attenuation at least $A_s = 40$ dB

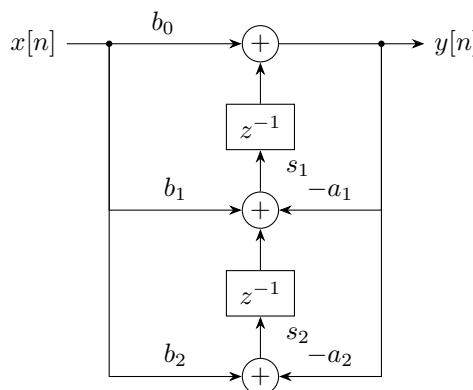
A biquad (second-order section) is written

$$H_k(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}},$$

a cascade is $H(z) = \prod_k H_k(z)$. Two structures are used: *direct form I* (DF I, states $x[n-1], x[n-2], y[n-1], y[n-2]$, one accumulator – preferred in fixed point) and the *transposed direct form II* (TDF II, two states s_1, s_2 – preferred in floating point):

$$y = b_0 x + s_1, \quad s_1 \leftarrow b_1 x - a_1 y + s_2, \quad s_2 \leftarrow b_2 x - a_2 y.$$

$Q15$ denotes a 16-bit two’s complement number $v = q/2^{15} \in [-1, 1)$, $Q2.14$ a 16-bit number $v = q/2^{14} \in [-2, 2)$.



Biquad in transposed direct form II.

Exercise 10.1 Design by hand**PAPER**

- Sampling rate.* The engine-sim chip updates its analog outputs every $20\text{ }\mu\text{s}$ (50 kHz, zero-order hold). Explain why the ESP32 samples the knock signal at $f_s = 25\text{ kHz}$ and not, e.g., at 40 kHz. Estimate the worst-case phase error of the 6.5 kHz component caused by the timing mismatch at 40 kHz. Is 25 kHz sufficient for both knock modes and for the valve disturbance? What happens to the white noise of the chip (generated at 50 kHz) when only every second value is sampled?
- FIR order.* Normalise the specification ($\Delta F = \Delta f/f_s$). Estimate the length N of a window-design FIR filter for a Hamming window ($N \approx 3.3/\Delta F$) and with Kaiser's formula $N - 1 \approx (A_s - 8)/(2.285 \Delta\omega)$, $\beta = 0.5842(A_s - 21)^{0.4} + 0.07886(A_s - 21)$. Where do you put the cut-off frequencies of the ideal band-pass?
- Derive the impulse response $h_d[m]$ of the ideal (zero-phase) band-pass with cut-offs $\omega_{c1} < \omega_{c2}$, and state how the causal FIR filter $h[n]$, $n = 0, \dots, N - 1$, is obtained from it. What is its group delay in samples, in ms, and in degrees crank angle at 3000 rpm and 6000 rpm? Why does this matter for a knock measurement window of 10° – 70° after TDC?
- IIR order.* Pre-warp the band edges, $\Omega = 2f_s \tan(\pi f/f_s)$, and compute the stop-band frequency $\lambda_s = \min |\Omega_s^2 - \Omega_0^2|/(B \Omega_s)$ of the normalised low-pass prototype ($\Omega_0^2 = \Omega_{p1}\Omega_{p2}$, $B = \Omega_{p2} - \Omega_{p1}$). Which Butterworth prototype order n is needed, how many biquads does the band-pass have, and how many multiplications per sample does it cost compared with the FIR filter?
- Second-order band-pass.* The knock burst decays with $\tau = 0.8\text{ ms}$. Show that its spectrum has a -3 dB bandwidth of $B_k = 1/(\pi\tau)$ and argue why a band-pass with about this bandwidth is a good compromise (matched filter). Design $H(s) = Bs/(s^2 + Bs + \Omega_0^2)$ with centre $f_0 = 6.5\text{ kHz}$ and digital bandwidth 400 Hz via the bilinear transform $s = 2f_s(1 - z^{-1})/(1 + z^{-1})$ with pre-warping (use $B \approx 2\pi \cdot 400\text{ Hz}/\cos^2(\pi f_0/f_s)$). Compute b_0, b_1, b_2, a_1, a_2 , the pole radius r and the decay time of the filter's impulse response.

Exercise 10.2 Filter design tool**JSLINUX**

Files in `code/jslinux/sheet10/`: `filtldesign.h` (design library), `filterdesign.c` (design, check, export), `coefquant.c` (word-length study), `filtertest.c` (test on the virtual engine).

```
1 gcc -O2 -o filterdesign filterdesign.c -lm && ./filterdesign # writes knock_filter.h
2 gcc -O2 -o coefquant coefquant.c -lm && ./coefquant && ./coefquant 200000
3 gcc -O2 -o filtertest filtertest.c -lm && ./filtertest
```

- FIR window method.* Complete `fd_window()` (Hann, Hamming, Blackman, Kaiser) and `fd_fir_bandpass()`. `filterdesign` prints, for $N = 83$ and every window, the worst pass-band loss and stop-band gain, and the smallest odd N that meets the specification. Explain the ranking of the windows. Is the Kaiser estimate of 10.1 b) met?
- IIR Butterworth band-pass.* Complete `fd_prewarp()` and the pole transformation in `fd_butter_bandpass()`: every prototype pole p gives the band-pass poles $s = \frac{1}{2}(pB \pm \sqrt{p^2 B^2 - 4\Omega_0^2})$, which are mapped with $z = (2f_s + s)/(2f_s - s)$. Check the specification, list the pole radii and frequencies of the sections, and compare the group delay at 6.5 kHz with the FIR filter. What goes wrong without pre-warping?
- Complete `fd_bp2()` and compare with your hand calculation of 10.1 e). Report the attenuation of the 400-Hz and a 1000-Hz version at 8 kHz and 10.5 kHz.
- Export.* Complete the Q15/Q2.14 conversion (rounding, saturation) and generate `knock_filter.h`. Why are the biquad coefficients exported in Q2.14 rather than in Q15? Why are the sections scaled to a peak gain of 1 (L_∞ scaling) and ordered by pole radius?

- e) *Coefficient quantisation.* Complete `quant_vec()` and the quantisation of a_1, a_2 in `coefquant.c`. The program realises the filter (1) as a biquad cascade and (2) as one 10th-order direct-form polynomial, quantises the coefficients to $B = 8 \dots 32$ bits and reports the pole displacement, the largest pole radius, the pass-band deviation and the stop-band gain. Interpret the results for $f_s = 25$ kHz and for the same analog specification at $f_s = 200$ kHz. Which word length do you need for the cascade, which for the direct form?
- f) *Test on the engine signal.* Complete `sos_step()` (TDF II, float) and `sosq_step()` (DF I, Q15 data, Q2.14 coefficients, 32-bit accumulator, rounding, saturation) in `filtertest.c`. Compare the window energies with and without knock for FIR and IIR, the delay of the filtered burst, and the error of the Q15 implementation.

Exercise 10.3 Real-time filtering on the microcontroller

WOKWI

- a) *ESP32, float.* Open `code/wokwi/sheet10_esp32_filter/` and replace the placeholder `knock_filter.h` by your generated header. The sketch samples the KNOCK pin at 25 kHz (deadline loop) and records 300 samples after each TDC of cylinder 1 (TDC pulse while CAM is HIGH). Complete `fir_step()` (circular buffer stored twice, so the newest N samples are always contiguous) and `iir_step()` (TDF II).
- Run at 3000 rpm with knock probability 100 % and 0 %. Copy one record into JSLinux and plot x , y_{FIR} , y_{IIR} and the KNK flag. Compare the window energies printed in the `# rec` lines.
 - Repeat at 6000 rpm. Explain the drastic change of the FIR window energy and propose a fix.
 - Why does the sketch use `float` and not `double` on the ESP32? Compare the indicative benchmark of FIR and IIR with the operation counts.
- b) *Uno, Q15.* Open `code/wokwi/sheet10_uno_q15/`. Timer1 interrupts at 25 kHz; the ISR reads the ADC, starts the next conversion and filters with the single biquad `kf_bp2_q14`.
- Complete `biquad_q15()`: DF I, 32-bit accumulator (Q29), rounding, conversion to Q15, overflow detection, saturation or wrap-around (`SATURATE`). Verify it in `MODE 0`, where a recorded block is filtered with float and with Q15 arithmetic.
 - Cycle budget: how many CPU cycles are available per sample? Measure the ISR duration (column `isr_cycles`, Timer1 count at the end of the ISR, or the pulse width on D8 with the logic analyzer). How many biquads, how many FIR taps would fit? Could the 10th-order cascade of 10.2 run on the Uno at 25 kHz?
 - Scaling: the input is mapped by $x = (\text{adc} - 512) \cdot 2^{\text{SHIFT}}$. Which `SHIFT` uses the full Q15 range? Show that the band-pass output can still overflow when the *input* clips (hint: a clipped sine approaches a square wave). Run `MODE 1` with knock amplitude 3 V and `SHIFT = 6, 8, 10` and compare saturation with wrap-around.
 - Complete the window energy `energy += y*y >> 8`. Show that the 32-bit accumulator cannot overflow down to 600 rpm. Why is $y^2 \gg 8$ much cheaper on the AVR than $y^2 \gg 12$?

Deliverables

10.1 on paper. `knock_filter.h` and the output of `filterdesign` and `coefquant` (both sampling rates); plot of one ESP32 record (x , y_{FIR} , y_{IIR} , KNK) at 3000 and 6000 rpm; Uno: block-test error, measured ISR cycles, table of overflow events for the three shifts with a short discussion.